

Chapter 1

Course Overview

- Columns, rows, matrices, $\mathbb{R}^n$, $\mathbb{R}^{m \times n}$ are reserved for representational uses.
- These represent respectively,
vectors, adjoint vectors, linear transformations, a real vector space X of dimensions n , $B(X, Y)$ between X and Y .
- As seen above, in this course, it is almost exclusively the case of a vector space over the real field $\mathbb{R}$.
 1. It is necessary to understand the geometric point of view for $\mathbb{R}^{m \times n}$, which might be obscured by using $\mathbb{C}$ field.
 2. Linear algebra over $\mathbb{R}$ is in general more complicated to study than that over $\mathbb{C}$, and once we are done with $\mathbb{R}$ field, the corresponding material with $\mathbb{C}$ is usually immediate, but not in the other way around.
 3. We know linear algebra over finite field $\mathbb{Z}_p$ is useful in some mathematical areas. We will not cover them unfortunately.

- A vector space X can be of various kinds, that of colors, tastes, quantum states, polynomials, $\dots$, where the scalar multiple and addition make senses.

For our expositional purpose, we mostly treat X as a vector space of

{Locations with respect to a fixed origin} in n -dimensional physical spaces.

1. $n = 3$ case will be most intuitive for this.
2. A location with a fixed origin can be scalar multiplied, and two locations with a fixed origin can be added by the parallelogram law.
3. Once we attach the coordinate basis to the set of locations, a location can be specified by its coordinates, listed in a column.

$$\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n.$$

Orthogonal Matrices

- We will call a matrix $O \in \mathbb{R}^{m \times n}$ orthogonal if columns of O are orthonormal.
- We give a definition that makes sense even when O is a rectangular matrix. If O is orthogonal, it is necessarily $m \geq n$.
- For a square orthogonal matrix $Q \in \mathbb{R}^{n \times n}$ consisting of n orthonormal columns $q_1, q_2, \dots, q_n$, it is obvious that

$$\begin{pmatrix} \text{---} & q_1^T & \text{---} \\ \text{---} & q_2^T & \text{---} \\ & \vdots & \\ \text{---} & q_n^T & \text{---} \end{pmatrix} \begin{pmatrix} | & | & \cdots & | \\ q_1 & q_2 & \cdots & q_n \\ | & | & \cdots & | \end{pmatrix} = \mathbf{I}.$$

In other words, Q^T is the inverse matrix of Q .

New representation after changing basis

Let $\mathbf{X}$ be the n -dimensional space of locations with fixed origin.

We will refer to the current coordinate basis

$$e_1, e_2, \dots, e_n, \quad e_i = (0, 0, \dots, 0, 1, 0, \dots, 0)^T \quad 1 \text{ on } i\text{-th position}$$

as *native basis*, while some other basis elements in $\mathbf{X}$

$$f_1, f_2, \dots, f_n$$

as *foreign basis*.

A location $x \in \mathbf{X}$ is currently represented by column

$$c = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

We want to know the new column whose entries represents the location x but in foreign basis

$$c' = \begin{pmatrix} x'_1 \\ x'_2 \\ \vdots \\ x'_n \end{pmatrix}.$$

To do this, (since it will be linear), we look for the native-to-foreign dictionary matrix $\square$ such that

$$c' = \square c$$

In many occasions, we are given foreign-to-native dictionary instead, that are

$$f_1, f_2, \dots, f_n \quad \text{represented in native basis.}$$

In other words, the listing of foreign basis in columns,

$$F = \begin{pmatrix} | & | & \cdots & | \\ f_1 & f_2 & \cdots & f_n \\ | & | & \cdots & | \end{pmatrix}$$

is telling us in native language to a foreign word c' : $Fc' = x$ is in native language that is c .

$$Fc' = c.$$

Hence, to get c' , the native-to-foreign dictionary matrix $\square = F^{-1}$.

In our numerical linear algebra course, the most important base are orthonormal ones. Thus,

$$c' = Q^T c$$

explains the procedure obtaining the new representation after changing basis that are (orthonormal) columns of Q .

Rectangular Matrices

- A matrix $A \in \mathbb{R}^{m \times n}$ represents a linear mapping A (denoted without having another symbol)

$$A : X \rightarrow Y,$$

a member of $B(X, Y)$ from X to Y of dimensions respectively n and m .

- There will be two parts of theory on rectangular matrices.
 1. $x \mapsto y = Ax$, the question is: what does A do on x to result in y ?
 2. From y to the solution x . We will completely specify the set of all least square solutions.

The first part

Let A be any matrix. $A \in \mathbb{R}^{m \times n}$. To understand what does A do, we first of all wish to characterize the followings.

- The kernel of A

$$\text{Ker}(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}.$$

- The range of A

$$\text{Ran}(A) = \{y \in \mathbb{R}^m \mid y = Ax \text{ for some } x \in \mathbb{R}^n\}.$$

Considering the matrix-column multiplication

$$\begin{pmatrix} | & | & | & \cdots & | \\ c_1 & c_2 & c_3 & \cdots & c_n \\ | & | & | & \cdots & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1 c_1 + x_2 c_2 + \cdots x_n c_n$$

we notice that

$$\text{Ran}(A) = \text{span} \langle c_1, c_2, \dots, c_n \rangle.$$

- The range of A^T , the transpose of A

$$\text{Ran}(A^T) = \{x \in \mathbb{R}^n \mid x = A^T y \text{ for some } y \in \mathbb{R}^m\}.$$

If we write

$$A = \begin{pmatrix} \text{---} & r_1^T & \text{---} \\ \text{---} & r_2^T & \text{---} \\ & \vdots & \\ \text{---} & r_m^T & \text{---} \end{pmatrix} \quad \text{then} \quad \text{Ran}(A^T) = \text{span} \langle r_1, r_2, \dots, r_m \rangle.$$

- The kernel of A^T

$$\text{Ker}(A^T) = \{y \in \mathbb{R}^m \mid A^T y = 0\}.$$

One immediately sees that

$$\begin{aligned} x \in \text{Ker}(A) & \text{ if and only if } x \perp \text{span} \langle r_1, r_2, \dots, r_m \rangle. \\ y \in \text{Ker}(A^T) & \text{ if and only if } y \perp \text{span} \langle c_1, c_2, \dots, c_n \rangle. \end{aligned}$$

In other words,

$$\mathbb{R}^n = \text{Ker}(A) \oplus \text{Ran}(A^T), \quad \mathbb{R}^m = \text{Ker}(A^T) \oplus \text{Ran}(A)$$

and they are orthogonal direct sums.

Fundamental Theorem of Linear Algebra:

Theorem 1. For any matrix $A \in \mathbb{R}^{m \times n}$,

$$\dim \text{span} \langle c_1, c_2, \dots, c_n \rangle = \dim \text{span} \langle r_1, r_2, \dots, r_m \rangle.$$

The equality of the two numbers gives rise to the definition of the rank.

- The rank of A

$$\text{rank}(A).$$

- $\text{rank}(A)$, $\text{Ker}(A)$, $\text{Ran}(A)$, $\text{Ker}(A^T)$, and $\text{Ran}(A^T)$ highlights about what happens in $x \mapsto Ax$.
- These will be studied in this course by
 1. QR decomposition
 2. Singular Value Decomposition.

The second part

The rootfinding part, that is to specify the complete set of all least square solutions, will be clear once we achieve QR -decomposition and Singular Value Decomposition.

Square Matrices

- In principle, all the materials for rectangular matrices applies to square matrices, where the vector spaces X and Y happen to have same dimensions.
- Inverse A^{-1} if exists, the determinant of A , $\dots$ are additionally discussed.
- The particular part for square matrices are when we identify the domain space and the range space, that is X . $A \in \mathbb{R}^{n \times n}$ represents

$$A : X \rightarrow X$$

In fancy words, A is an endomorphism, $A \in \text{End}(X)$.

1. Considering X as space of locations, $y = Ax$ is the new location in X referenced by $x \in X$. Then we can speak of shape deformation from the referenced shape $G \subset X$ by A .
2. We can then speak of rotation, reflection, dilation, $\dots$.
3. Most important concept in $\text{End}(X)$ is the invariance under A . We look for $M \subset X$ such that AM does not escape M .

Invariant vector subspace under A

- In the simplest setting, this is the eigenvalue problem.
- We will treat two cases:
 1. Diagonalization for a symmetric matrix A .
 2. Schur decomposition for any square matrix A .

Today, we examine the fact and its consequences that

- there exists a stably-working numerical algorithm implementing
- diagonalization of a real square symmetric matrix.

1. We would like to talk about why this is a nontrivial fact.
2. Believing the fact, we show we will be able to, not only define but also compute

$\text{rank}, \text{Ran}, \text{Ker}, \dots$

for any $m \times n$ matrix.

Theorem. *For a $n \times n$ real symmetric matrix A , there exists an orthogonal matrix $O \in \mathbb{R}^{n \times n}$ and a diagonal matrix $\Lambda \in \mathbb{R}^{n \times n}$ such that*

$$A = O\Lambda O^T.$$

Singular Value Decomposition

- Here, we give a proof of the existence of Singular Value Decomposition for any matrix $A \in \mathbb{R}^{m \times n}$.
- The point of presenting this proof is to see if we can build an algorithm for SVD.

Theorem. *For any matrix $A \in \mathbb{R}^{m \times n}$, there exists a matrix $O \in \mathbb{R}^{m \times m}$, $\Sigma \in \mathbb{R}^{m \times n}$, and $Q \in \mathbb{R}^{n \times n}$ such that*

1. O is an orthogonal matrix,
2. Q is an orthogonal matrix,
3. Σ is such that if $k = \min\{m, n\}$, the top left maximal $k \times k$ block is diagonal with diagonal entries

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_k \geq 0$$

and all other entries of Σ are zero,

4. $A = O\Sigma Q^T$.

Proof. .

1. We may give a proof for the case $m \geq n$. If not, we achieve

$$A^T = O\Sigma Q^T \implies A = Q\Sigma^T O^T$$

which is in the form as stated. Assume now $m \geq n$.

2. We consider a $n \times n$ symmetric matrix $A^T A$. We first check that

$$A^T A x = 0 \iff A x = 0.$$

This is because

$$A^T A x = 0 \implies x^T A^T A x = 0 \implies |Ax|^2 = 0 \implies Ax = 0.$$

Converse is easy.

3. $A^T A$ is a real symmetric and nonnegative definite. Hence, we have

$$A^T A = Q\Lambda Q^T.$$

The diagonalization can be arranged so that diagonal entries of Λ are in descending order:

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n \geq 0.$$

We let $r \leq n$ be the number such that

$$\lambda_r > 0 \quad \text{and} \quad i > r \implies \sigma_i = 0. \quad \text{Set } r = 0 \text{ if all eigenvalues are zero.}$$

and let $I_r = \{1, 2, \dots, r\}$. The set may be empty if $r = 0$.

In particular, if $\lambda_i = 0$ then

$$\lambda_i q_i = A^T A q_i = 0 \implies A q_i = 0.$$

4. Now, define $\mathbb{R}^m$ -vectors for $i \in I_r$:

$$\text{for } i \in I_r \quad o_i = \frac{Aq_i}{\sqrt{\lambda_i}}.$$

We check

$$\text{for } i, j \in I_r, \quad o_i^T o_j = \frac{q_i^T A^T A q_j}{\sqrt{\lambda_i \lambda_j}} = \frac{\lambda_j q_i^T q_j}{\sqrt{\lambda_i \lambda_j}} = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

5. If $\{r+1, r+2, \dots, m\}$ is nonempty, supplement $o_{r+1}, o_{r+2}, \dots, o_m$ so that

$$o_1, o_2, \dots, o_m \quad \text{are orthonormal.}$$

6. Let O, Q, Σ be such that: O consists of columns $o_1, o_2, \dots, o_m$, Q consists of columns $q_1, q_2, \dots, q_n$, and Σ is as stated in Theorem with the top left diagonals

$$\sigma_i = \sqrt{\lambda_i}, \quad i \in I_r, \quad \text{and define } \sigma_{r+1} = \sigma_{r+2} = \dots = \sigma_n = 0.$$

Finally, we show that

$$Ax = O\Sigma Q^T x \quad \text{for every } x \in \mathbb{R}^n.$$

Since $q_1, q_2, \dots, q_n$ forms a basis of $\mathbb{R}^n$, let us suppose

$$x = c_1 q_1 + c_2 q_2 + \dots + c_n q_n.$$

In other words, $c = Q^T x$. Then

$$\begin{aligned} Ax &= \sum_{i=1}^n c_i Aq_i \\ &= \sum_{i \in I_r} c_i Aq_i && (Aq_i = 0 \text{ for } i > r) \\ &= \sum_{i \in I_r} (\sigma_i c_i) o_i && (\text{definition of } o_i) \\ &= \sum_{i=1}^n (\sigma_i c_i) o_i && (\sigma_i = 0 \text{ for } i > r) \\ &= \sum_{i=1}^n (\Sigma Q^T x)_i o_i \\ &= \sum_{i=1}^m (\Sigma Q^T x)_i o_i && (\text{The component } (\Sigma Q^T x)_i = 0 \text{ for } i > n) \\ &= O\Sigma Q^T x. \end{aligned}$$

□

For a given any matrix $A \in \mathbb{R}^{m \times n}$, we are given a Singular Value Decomposition

$$A = O\Sigma Q^T.$$

Then we can compute

- $\text{rank}(A) = r$, where r is the number of nonzero singular values. If singular values are in descending order,

$$\sigma_r > 0 \quad \text{and} \quad i > r \implies \sigma_i = 0.$$

- $\text{Ker}(A) = \text{span} \langle q_{r+1}, q_{r+2}, \dots, q_n \rangle.$
- $\text{Ran}(A) = \text{span} \langle o_1, o_2, \dots, o_r \rangle.$
- $\text{Ran}(A^T) = \text{span} \langle q_1, q_2, \dots, q_r \rangle.$
- $\text{Ker}(A^T) = \text{span} \langle o_{r+1}, o_{r+2}, \dots, o_m \rangle.$

These computations contrast their definitions

- $\text{rank}(A)$, the number of independent columns (= number of independent rows)
- $\text{Ker}(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}.$
- $\text{Ran}(A) = \{y \in \mathbb{R}^m \mid y = Ax \text{ for some } x \in \mathbb{R}^n\}.$
- and those for A^T .

Examine the steps 1-6 of the proof of the Singular Value Decomposition. Will we be able to build an algorithm?

It will still be a long journey until we achieve an algorithm for the diagonalization of a real symmetric square matrix.

Chapter 2

Square Matrices 1 : Elementary Square Matrices

Today, we begin Square Matrices 1.

- In general,
 $A : X \rightarrow Y$ is represented by a rectangular matrix.
- In this section, we consider a special case

$A : X \rightarrow X$ is represented by a square matrix.

In a proper words, $A \in \text{End}(X)$.

To be concrete, we mostly consider the $n = 3$ case. One can consider $A \in \mathbb{R}^{3 \times 3}$ as a geometric linear deformation of a shape.

- The point of speaking of elementary square matrices is that we do not need to know many kinds. There are only 5 kinds of linear deformation, and a general square matrix is a finite compositions of the 5 kinds.

We only need to know 5 kinds of elementary types which we will study in this section.

1. dilation,
 2. rotation,
 3. reflection,
 4. shearing,
 5. projection.
- You will build codes implementing the (square matrix-column) multiplication, in case the matrix is elementary. These codes will be building blocks for the algorithms in this course.

Very elementary square matrices $\rightarrow$ elementary square matrices $\rightarrow$ general square matrices.

Very elementary square matrices

These are elementary ones, and even based on the coordinate axes.

1. Dilation

We consider matrices of the form

$$\text{the identity matrix } I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and the diagonal matrix } \Lambda = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix}$$

For the moment, let us consider cases $\lambda_1, \lambda_2, \lambda_3$ are strictly positive.

This describes dilation with respect to x , y , and z axis by scale factors of λ_1 , λ_2 , and λ_3 .

2. Reflection with respect to $n - 1$ dimensional hyperplane

In 3-dimensions, there are 3 very elementary reflections, that are with respect to yz -plane, zx -plane, and xy -plane:

$$\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

3. Orthogonal projections onto k -dimensional planes

These forms very elementary projection matrices

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \dots$$

onto xy -plane, x -axis, yz -plane, y -axis, $\dots$.

4. Shearing

In 3-dimensions, there are very elementary shearings of the form

$$\begin{pmatrix} 1 & t & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & t \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & t \\ 0 & 0 & 1 \end{pmatrix}$$

and in general an upper triangular matrix

$$\begin{pmatrix} \lambda_1 & t & r \\ 0 & \lambda_2 & s \\ 0 & 0 & \lambda_3 \end{pmatrix}.$$

- Shearing is a deformation that takes place with respect to cascading pivot axes:
 $x\text{-axis} \rightarrow xy\text{-plane} \rightarrow xyz\text{-space} \rightarrow \dots$

5. Rotation of 2-diminsinal subspaces keeping others invariant

In 3-dimensions, there are 3 very elementary rotations

$$\begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}, \quad \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$$

Generalizations into $\mathbb{R}^n$ cases can also be considered:

Elementary square matrices

- We have seen 5 kinds that are based on coordinate axes.
- Combining these with basis changing technique, we come up with new 5 kinds:
 1. If U is a very elementary square matrix and
 2. $Q \in \mathbb{R}^{n \times n}$ is the relevant orthogonal matrix, then

$$T = QUQ^T \quad \text{is an elementary square matrix}$$

3. based on new axes specified by columns of Q .
- We will examine one-by-one though below.
 - We do this in the order: projection, reflection, rotation, dilation, shearings.

Orthogonal projections onto k -dimensional planes

- Suppose an orthogonal matrix Q is given in such a way that
 1. $\text{span} \langle q_1, q_2, \dots, q_k \rangle$ is the k -dimensional plane to project on.
 2. $\text{span} \langle q_{k+1}, q_{k+2}, \dots, q_n \rangle$ is the complementary orthogonal plane.
- Then the corresponding projection matrix ($n = 3, k = 2$ case)

$$T = Q \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} Q^T.$$

- In fact, if we only specify k vectors $q_1, q_2, \dots, q_k$,

$$Q \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} Q^T = \left\{ \begin{pmatrix} | & | \\ q_1 & q_2 \\ | & | \end{pmatrix} \begin{pmatrix} \text{---} & q_1^T & \text{---} \\ \text{---} & q_2^T & \text{---} \end{pmatrix} \right\} = \hat{Q}\hat{Q}^T, \quad \hat{Q} = \begin{pmatrix} | & | \\ q_1 & q_2 \\ | & | \end{pmatrix}$$

Reflection with respect to $n - 1$ dimensional hyperplane

- Suppose an orthogonal matrix Q is given in such a way that
 1. q_1 is unit normal to a certain $(n - 1)$ -dimensional hyperplane, and
 2. $\text{span}\langle q_2, q_3, \dots, q_n \rangle$ is the hyperplane.
- Then the corresponding reflection matrix ($n = 3$ case)

$$T = Q \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} Q^T.$$

- In fact, if we only specify the normal vector q_1 ,

$$\begin{aligned} Q \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} Q^T &= Q \left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} -2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\} Q^T = QQ^T - 2q_1q_1^T \\ &= I - 2q_1q_1^T. \end{aligned}$$

Rotation of 2-dimensional subspaces keeping others invariant

- Suppose an orthogonal matrix Q is given in such a way that
 1. $\text{span}\langle q_1, q_2 \rangle$ is the 2-dimensional plane to rotate counterclockwise, and
 2. $\text{span}\langle q_3, q_4, \dots, q_n \rangle$ are to be kept invariant.
- Then the corresponding rotation matrix ($n = 3$ case)

$$T = Q \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} Q^T.$$

- In fact, if we only specify q_1 and q_2 ,

$$\begin{aligned} Q \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} Q^T &= Q \left\{ \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right\} Q^T \\ &= \left\{ \hat{Q} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \hat{Q}^T \right\} + I - \hat{Q}\hat{Q}^T, \quad \hat{Q} = \begin{pmatrix} | & | \\ q_1 & q_2 \\ | & | \end{pmatrix} \end{aligned}$$

Dilation

- Suppose an orthogonal matrix Q is given in such a way that each q_i is axis of dialation.
- Then the corresponding dialation matrix is

$$T = Q \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix} Q^T$$

- Matrices made in this way is symmetric.
- Conversely, according to Linear Algebra theory, any symmetric matrix can be put in the above form.

Shearings

- Suppose an orthogonal matrix Q is given in such a way that the cascading axes are
 q_1 -axis $\rightarrow q_1 q_2$ -plane $\rightarrow q_1 q_2 q_3$ -space $\rightarrow \dots$
- Then the corresponding shearing matrix is

$$T = Q \begin{pmatrix} \lambda_1 & t & r \\ 0 & \lambda_2 & s \\ 0 & 0 & \lambda_3 \end{pmatrix} Q^T$$

- Matrices made in this way is characterized by a square matrix whose eigenvalues are all real.
- Conversely, according to Linear Algebra theory, any such a matrix can be put in the above form.

Home Work 1: Important for later algorithms

1. Implement functions by yourself the elementary projection, reflection, and rotations.
 - Input: x and $q_1, q_2, \dots$ that are relevant.
 - Output: y

Remark 2.1. .

- Those converse statements for dilation and shearing are subject of eigenvalues, invariant subspaces, ... This will be treated in the last part of the course.
- We can state the following fact for square matrices though in advance.
- Let U be a very elementary square matrix, and consider T that can be put into

$$T = QUQ^T, \quad \underline{Q \text{ is orthogonal.}}$$

- The converse statements go in the correspondence below:

$$\begin{array}{llll} T : & \{\text{symmetric matrices}\} & \subset & \{\text{matrices with real eigenvalues}\} & \subset & \{\text{all matrices}\} \\ U : & \{\text{diagonal matrices}\} & \subset & \{\text{upper triangular matrices}\} & \subset & \{\text{schur form matrices}\} \end{array}$$

Home Work 2:

1. Verify that elementary reflection and elementary rotation matrices are orthogonal.

Remark 2.2. .

- Elementary reflection and elementary rotation matrices are orthogonal. The former has determinant -1 while latter has determinant $+1$.
- Any orthogonal matrix with determinant $+1$ can be made by finite times compositions of very elementary rotations.
- Any orthogonal matrix with determinant -1 can be made by above followed by one very elementary reflection.

Chapter 3

Rectangular Matrices 1

QR decomposition

- We are equipped with the elementary reflection, rotation, and projection.
- Then, we can do much stronger achievement, that is the QR decomposition.
- We very quickly achieve this, from one perspective of QR among many interpretations: we monitor $\text{Ran}(A^T)$ while $\text{Ran}(A)$ incidently follows at the end.
- After this quick achievement, we review material around QR from the beginning again,
 1. other perspective where we monitor $\text{Ran}(A)$ while $\text{Ran}(A^T)$ incidently follows,
 2. meaning of them, numerical perspective too, and so on.

We recall matrix manipulation we have seen in Gauss Elimination.

- We say, $A, A', A'', \dots$ are *Row equivalent* if

$$\text{Ran}(A^T) = \text{Ran}(A'^T) = \text{Ran}(A''^T) = \dots$$

- The objective of Gauss Elimination is to come up with
 1. simpler and simpler matrices but row equivalent.
In most of textbook examples, this is understood as a job for human-students.
 2. Typically one turns A into R-form (defined below), further to Row Echelon Form, and finally to Reduced Row Echelon Form.

Definition. A matrix $A \in \mathbb{R}^{m \times n}$ is in *R-form* if

$$A_{ij} = 0 \quad \text{for } i > j.$$

- Let us for later purpose consider following: $A, A', A'', \dots$ are *Column equivalent* if

$$\text{Ran}(A) = \text{Ran}(A') = \text{Ran}(A'') = \dots$$

Now, we give a statement of QR decomposition in the most general form.

Theorem 1. *For any $A \in \mathbb{R}^{m \times n}$, there exist $Q \in \mathbb{R}^{m \times m}$ and $R \in \mathbb{R}^{m \times n}$ such that*

1. Q is orthogonal.
2. R is in Row Echelon Form.
3. $A = QR$.

That we achieve Row Echelon Form includes that in the intermediate steps we identify what entries are pivots in R .

Apparent consequences of Theorem 1 in exact arithmetics

- Suppose we achieved

$$A = QR, \quad Q \text{ orthogonal, and } R \text{ REF.}$$

- This is compared to another REF produce by Gauss Elimination, where we write

$$A = PLU, \quad U \text{ is REF, } L \text{ is lower triangular, } P \text{ is a permutation.}$$

- We can read off immediately:
 1. $\text{rank}(A)$ is the number of pivots in REF.
 2. $\text{Ran}(A) = \text{span} \langle \text{pivot columns of } Q \rangle$
 3. $\text{Ran}(A^T) = \text{span} \langle \rho_1, \rho_2, \dots, \rho_r \rangle$ of nonzero rows of R . ρ_i is the i -th row of the REF R .
- Unlike the SVD, situation for the range space Y is certainly better than that of the domain space X .
 1. The pivot columns $q_{i(1)}, q_{i(2)}, \dots, q_{i(r)}$ are orthonormal. The complementary columns of Q spans $\text{Ker}(A^T)$.
 2. The pivot rows $\rho_1, \rho_2, \dots, \rho_r$ are as in REF. To characterize $\text{Ker}(A) = \text{Ker}(R)$, we have to collect all solutions of

$$Rx = 0$$

- To determine a certain column is whether a pivot column or not is numerically not as perfect as we want. We can say **Revealing strength** in machine arithmetics, not in exact arithmetics, for $\text{rank}(A)$, $\text{Ran}(A)$, $\text{Ran}(A^T)$ is as good as that of PLU decomposition.
- This is not as good as we want to achieve, but can be practical in many situations.

Very quick illustration for the algorithm for Theorem 1

- Either using the elementary rotations
- or using the elementary reflections

by pictures.

If we take the simpler algorithm omitting the step deciding whether a certain column is a pivot column or not, we achieve mere R-form:

Theorem 2. *For any $A \in \mathbb{R}^{m \times n}$, there exist $Q \in \mathbb{R}^{m \times m}$ and $R \in \mathbb{R}^{m \times n}$ such that*

1. Q is orthogonal.
2. R is in R-Form.
3. $A = QR$.

If R is merely in R-form, not in REF, in general we cannot read off $\text{Ran}(A)$ and $\text{Ran}(A^T)$ out of

$$A = QR.$$

- Example:

$$A = \begin{pmatrix} | & | & | \\ q_1 & q_2 & q_3 \\ | & | & | \end{pmatrix} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} | & | & | \\ q_1 & q_1 & q_1 + q_2 + q_3 \\ | & | & | \end{pmatrix}$$

$\text{rank}(A) = 2$, but how to compose $\text{Ran}(A^T)$ and $\text{Ran}(A)$ of 2-dimensions?

- If so, then for what purposes do we do this decomposition in Theorem 2?
- We achieve much less, in general, by Theorem 2. The objective of doing this will be specified when we review another perspective of QR decomposition: $\text{Ran}(A)$ -monitoring one.

Home Work :

1. Implement algorithms for Theorem 1 (with pivot identification) and Theorem 2 (without pivot identification)
2. Input : any matrix $A \in \mathbb{R}^{m \times n}$
3. Output : (Q, R) , where $Q \in \mathbb{R}^{m \times m}$ orthogonal, $R \in \mathbb{R}^{m \times n}$ either REF or R-form.

Before we go further, we set the following result as our finest QR decomposition. This is called the QR decomposition + Column pivoting. (qrcp)

Theorem 3. *For any $A \in \mathbb{R}^{m \times n}$, there exist $O \in \mathbb{R}^{m \times m}$, $R \in \mathbb{R}^{m \times n}$, $Q \in \mathbb{R}^{n \times n}$ such that*

1. $A = ORQ^T$.
2. O is orthogonal.
3. Q is orthogonal, in particular it is a permutation.
4. R is in Row Echelon Form.

- We have said that **Revealing strength** in machine arithmetics for Theorem 1 may be practical.
- **Revealing strength** in machine arithmetics for Theorem 3 is as accurate as we want to achieve.

Home Work :

1. Implement algorithms for Theorem 3 with the column pivoting.

Remark 3.1. Comparing SVD ($A = O\Sigma Q^T$) to the decomposition in Theorem 3,

1. R is not as simple as the diagonal Σ in SVD while
2. Q is simpler here, that is just a permutation.

Review on QR decomposition

QR decomposition is the place where we can learn all the important concepts arising in numerical algorithms and their analysis.

More relevantly as a pedagogical purposes, we will visit one-by-one those concerns, and we will learn so many useful concepts.

Review 1. Same problem, Different perspective, Different algorithm

Of course, it is natural that there are many ways of resolving one same problem. QR decomposition example illustrates how choosing one particular way affects the later studies.

We now introduce one objective seemingly quite different from the one we had before:

- We turn $A \in \mathbb{R}^{m \times n}$ into $A', A'', \dots$ that are column equivalent to each other.
- In particular we wish to have a property

$$A = \begin{pmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{pmatrix}, \quad A' = \begin{pmatrix} | & | & \cdots & | \\ a'_1 & a'_2 & \cdots & a'_n \\ | & | & \cdots & | \end{pmatrix}$$

so that

$$\begin{aligned} \text{span} \langle a_1 \rangle &= \text{span} \langle a'_1 \rangle \\ \text{span} \langle a_1, a_2 \rangle &= \text{span} \langle a'_1, a'_2 \rangle \\ \text{span} \langle a_1, a_2, a_3 \rangle &= \text{span} \langle a'_1, a'_2, a'_3 \rangle \\ &\vdots \\ \text{span} \langle a_1, a_2, a_3, \dots, a_n \rangle &= \text{span} \langle a'_1, a'_2, a'_3, \dots, a'_n \rangle. \end{aligned}$$

If this is the case, and if all a'_j are orthonormal, this is to achieve the Gram-Schmidt Orthogonalization for a given family

$$a_1, a_2, \dots, a_n.$$

- Since the family might not be independent, we expect less:

$$\begin{aligned} \text{span} \langle a_1 \rangle &\subseteq \text{span} \langle a'_1 \rangle \\ \text{span} \langle a_1, a_2 \rangle &\subseteq \text{span} \langle a'_1, a'_2 \rangle \\ \text{span} \langle a_1, a_2, a_3 \rangle &\subseteq \text{span} \langle a'_1, a'_2, a'_3 \rangle \\ &\vdots \\ \text{span} \langle a_1, a_2, a_3, \dots, a_n \rangle &\subseteq \text{span} \langle a'_1, a'_2, a'_3, \dots, a'_n \rangle. \end{aligned}$$

This amounts to say that a_j is a linear combination of $a'_1, a'_2, \dots, a'_j$, and thus

$$A = \begin{pmatrix} | & | & \cdots & | \\ a_1 & a_2 & \cdots & a_n \\ | & | & \cdots & | \end{pmatrix} = \begin{pmatrix} | & | & \cdots & | \\ a'_1 & a'_2 & \cdots & a'_n \\ | & | & \cdots & | \end{pmatrix} R = A' R, \quad R \text{ is in R-form.}$$

- Note that, in this case it is natural to have: A' is $m \times n$, and R is $n \times n$.
(Most relevant cases are when $m \geq n$.)
- This objective and our previous one has some consequences in common.
- We will soon discuss that from numerical perspective, this way of achieving QR is inferior choice in many senses.

Review 2. QR decomposition is an ill-conditioned problem.

It is easy to illustrate that the procedure $A \mapsto (Q, R)$ such that $QR = A$ is ill-conditioned.

By picture:

1. For an ill-conditioned problem, one cannot expect to build an algorithm with a small solution error.
2. However, this does not mean that one cannot build an algorithm with a small testing error. We actually can.

Let us be precise in saying that. Let us define the problem $\mathbf{P}_{QR}$:

1. Let the map

$$F : \{Q \in \mathbb{R}^{m \times m} \mid Q \text{ is orthogonal}\} \times \{R \in \mathbb{R}^{m \times n} \mid R \text{ is in R-form}\} \rightarrow \mathbb{R}^{m \times n}$$

$$F(Q, R) = QR,$$

2. We let $\mathbf{X} = \mathbf{X}_1 \times \mathbf{X}_2$, $\mathbf{Y} = \mathbb{R}^{m \times n}$

$$\mathbf{X}_1 = \{Q \in \mathbb{R}^{m \times m} \mid Q \text{ is orthogonal}\}, \quad \mathbf{X}_2 = \{R \in \mathbb{R}^{m \times n} \mid R \text{ is in R-form}\}.$$

3. For a given $A \in \mathbf{Y}$, find a solution of $F(Q, R) = A$.

The second item amounts to specify objectives:

2-(a) for $k = \min\{m, n\}$

$$\begin{aligned} \text{span} \langle a_1 \rangle &\subseteq \text{span} \langle q_1 \rangle \\ \text{span} \langle a_1, a_2 \rangle &\subseteq \text{span} \langle q_1, q_2 \rangle \\ \text{span} \langle a_1, a_2, a_3 \rangle &\subseteq \text{span} \langle q_1, q_2, q_3 \rangle \\ &\vdots \\ \text{span} \langle a_1, a_2, a_3, \dots, a_k \rangle &\subseteq \text{span} \langle q_1, q_2, q_3, \dots, q_k \rangle. \end{aligned}$$

2-(b) $q_1, q_2, \dots, q_m$ are orthonormal.

We expect to be able to build an algorithm so that :

1. For given any $A \in \mathbb{R}^{m \times n}$, the algorithm H produces

$$H(A) = (\bar{Q}, \bar{R}).$$

2. The algorithm is consistent in the sense

(a) $(\bar{Q}, \bar{R}) \in \mathbf{X}$. That is to say, $\bar{Q}$ is orthogonal and $\bar{R}$ is in R-form.

(b) $\|A - \bar{Q}\bar{R}\| \leq \epsilon$

for an appropriate $\epsilon > 0$. (The matrix norm will be discussed in the next chapter.)

However, this does not say that $(\bar{Q}, \bar{R})$ is close to any of true decomposition (Q, R) of A .

Review 3. The way of the 2nd approach can numerically fail to be consistent.

We consider what can happen if we proceed with the 2nd approach in cases

1. without pivot identification.
2. with pivot identification.
3. with column pivoting.